PRODUCT LINE PROFITABILITY AND MARGIN PERFORMANCE ANALYSIS

A Data-Driven Analysis of Nassau Candy Distributor

Project research report
Prepared as part of a Data Science Internship Project

Dataset: Nassau Candy Distributor | 10,194 records | 15 unique products

This is a project report based on the supplied dataset and completed analysis. It is not represented as a peer-reviewed, approved, or published academic paper.

Abstract

This project examines product-line profitability and gross-margin performance using a transactional dataset for Nassau Candy Distributor. The workflow includes data inspection, preparation, financial validation, derived profitability measures, summary analysis, and an interactive Streamlit dashboard. The dataset contains 10,194 records and 18 original columns, representing 15 unique products. The analysis reports total revenue of $141,783.63, total cost of $48,340.83, and total gross profit of $93,442.80, resulting in an overall gross margin of 65.91%. Chocolate is the largest division by revenue and gross profit. The results demonstrate why product decisions should consider both absolute gross-profit contribution and gross-margin percentage.

Keywords: profitability analysis, gross margin, product performance, exploratory data analysis, Python, Streamlit.

1. Introduction

Profitability analysis helps organisations understand how effectively products convert sales into gross profit. Revenue alone does not show whether a product is commercially attractive. Comparing revenue, cost, gross profit, units sold, and gross margin provides a more balanced view of product and division performance. This project combines data analysis and dashboard development to support exploration of those measures.

2. Problem Statement

The project investigates which products and divisions contribute most to revenue and gross profit, how their gross margins compare, and how the results can be explored through an interactive dashboard.

3. Objectives

- Inspect dataset structure and quality.
- Validate sales, cost, and gross-profit relationships.
- Calculate gross margin, profit per unit, and contribution measures.
- Compare product and division performance.
- Explore monthly and regional patterns.
- Build a filterable dashboard to communicate findings.

4. Dataset Description

The source file, Nassau Candy Distributor.csv, contains 10,194 rows and 18 original columns. Fields include order and shipping dates, customer and geographic attributes, division, product identifiers and names, sales, units, gross profit, and cost. The analysis identifies 15 unique products.

Dataset characteristic	Value
Records	10,194
Original columns	18
Unique products	15
Missing values in original columns	0
Exact duplicate rows	0

5. Tools and Technologies

Python is used for data processing and analysis. Pandas and NumPy support tabular data handling and calculations. Streamlit provides the interactive web application, and Plotly is used for interactive charts. GitHub is used for project version control and source-code hosting.

6. Methodology

6.1 Data inspection and preparation

The workflow reads the CSV file, inspects dimensions and columns, checks missing values and duplicate records, and converts date fields for time-based analysis. Text fields are standardised. Quality checks flag missing required values, non-positive sales or units, negative costs or gross profit, and shipping dates earlier than order dates.

6.2 Financial validation and derived measures

Gross profit is checked against the relationship between sales and cost. Derived measures include calculated profit, profit difference, gross margin, profit per unit, revenue contribution, profit contribution, and order month. Cleaned records and summary tables are generated for division, product, and monthly analysis.

6.3 Dashboard development

The Streamlit dashboard includes date, division, and product filters, product text search, KPI cards, division comparisons, top products by gross profit, monthly trends, product margin comparisons, regional performance, and a downloadable filtered CSV.

7. Key Performance Indicators

Gross Profit = Sales Revenue – Cost
Gross Margin (%) = (Gross Profit ÷ Sales Revenue) × 100
Profit per Unit = Gross Profit ÷ Units Sold
Revenue Contribution (%) = (Product or Division Revenue ÷ Total Revenue) × 100
Profit Contribution (%) = (Product or Division Gross Profit ÷ Total Gross Profit) × 100

Gross profit is a monetary amount, while gross margin is a percentage. Gross margin represents the proportion of revenue remaining after the costs represented in the dataset; it is not net profit and does not necessarily include all operating expenses.

8. Results and Findings

Overall metric	Result
Total revenue	$141,783.63
Total cost	$48,340.83
Total gross profit	$93,442.80
Overall gross margin	65.91%
Valid records	10,194
Records flagged for review	0
Suspicious shipping dates	0

The implemented checks report no missing values in the original columns, no exact duplicate rows, no profit-validation mismatches, no records flagged for review, and no suspicious shipping dates. These results reflect the validation rules used and do not guarantee that every possible data issue has been detected.

8.1 Division performance

Division	Revenue ($)	Cost ($)	Gross profit ($)	Units	Margin
Chocolate	131,692.90	42,868.28	88,824.62	37,275	67.45%
Other	9,663.25	5,329.80	4,333.45	1,242	44.84%
Sugar	427.48	142.75	284.73	137	66.61%

Chocolate is the dominant division by revenue and gross profit. Other has the lowest gross margin (44.84%), suggesting a need for deeper product-level review. Sugar has a relatively high margin percentage but a small revenue and profit base.

8.2 Product-level observations

Product	Revenue ($)	Gross profit ($)	Margin
Wonka Bar - Scrumdiddlyumptious	27,874.80	19,357.50	69.44%
Wonka Bar - Triple Dazzle Caramel	28,485.00	18,610.20	65.33%
Wonka Bar - Milk Chocolate	26,867.75	17,443.37	64.92%
Wonka Bar - Nutty Crunch Surprise	23,574.95	16,819.95	71.35%
Wonka Bar - Fudge Mallows	24,890.40	16,593.60	66.67%
Everlasting Gobstopper	130.00	104.00	80.00%
Kazookles	1,205.75	92.75	7.69%

The five Wonka Bar products listed generate the largest gross-profit amounts in the product summary. Everlasting Gobstopper has an 80.00% margin in the results shown but only $130.00 revenue. Kazookles has a low margin of 7.69%, making it a candidate for review of pricing, cost, discounting, or product strategy. High margin on a small revenue base should not be mistaken for high total profit contribution.

9. Business Recommendations

• Monitor availability, cost changes, and pricing for Chocolate, the largest contributor to revenue and gross profit.
• Review the cost and selling-price structure of low-margin items such as Kazookles and the Other division.
• Evaluate both gross-margin percentage and gross-profit amount before changing product strategy.
• Monitor monthly sales and gross-profit trends as new data becomes available.
• Refresh the analysis regularly and compare consistent time periods.

10. Limitations

The analysis is limited to the fields and records in the supplied CSV. Gross profit and gross margin use the dataset's Sales, Cost, and Gross Profit fields and should not be interpreted as net profit. The analysis is descriptive and does not establish causality. External market conditions, competitor pricing, overheads, taxes, inventory availability, and other factors are not assessed unless represented in the supplied data. Findings should be validated with business stakeholders before operational decisions.

11. Conclusion

This project demonstrates an end-to-end workflow for analysing product and division profitability. Across 10,194 records and 15 unique products, the overall gross margin is 65.91%. Chocolate contributes the largest revenue and gross profit. Product-level results show the importance of considering both margin percentage and absolute gross-profit contribution. The Streamlit dashboard makes the analysis easier to explore and communicate.

12. Project Resources

GitHub repository: https://github.com/vengalluruchandhana-hub/Nassau-Candy-Profitability-Analysis
Live dashboard: https://profitability-insights-2026.streamlit.app

13. Data Source and References

Primary data source: Nassau Candy Distributor.csv, supplied for this project. Numerical findings are based on the completed project analysis. Technical references: Python (python.org), pandas (pandas.pydata.org), Streamlit (docs.streamlit.io), and Plotly (plotly.com/python).

Document status: This is an internship project research report. It has not been submitted to or accepted by an academic journal or external publication.